

### Blockchain Technology Questions:

1. Describe the working principle of Bitcoin Cryptocurrency or Bitcoin Blockchain technology.
2. What are the security goals? Briefly describe the attacks on each security goal.
3. A person wants to transfer 5 Ether to his friend. In total  $5 + (42 \times 10^{-4})$  amount is deducted from his account. If the given boundary is 21,000, find the number of gwei per unit gas.
4. Calculate the amount of ether required to transfer 1 ether from X to Y where max gas consumption boundary is 21,000 and 1 Gas = 200 gwei.
5. What are the different layers of Blockchain technology? Explain each component.
6. What is a hash function? Explain its properties and characteristics.
7. What is a Digital Signature? Explain its working and role in bitcoin and cryptocurrency.
8. Describe the process of Proof of Work in bitcoin cryptocurrency.
9. Describe the characteristics of centralized, decentralized and decentralized systems with their use cases.
10. What are the generations of web? Describe them.